

Machine Learning — Fundamentals

Question Paper

Name: _____ Time: 3 Hours
Roll No: _____ Max Marks: 100

General Instructions

- All questions are compulsory unless stated otherwise.
- Section A is objective; Sections B–D require written answers.
- Draw diagrams/equations wherever necessary. Scientific calculators are allowed.

Section A — Multiple Choice Questions

(10 × 1 = 10 Marks)

- Q1.** Which of the following is a supervised learning algorithm?
(a) K-Means (b) Linear Regression (c) PCA (d) DBSCAN
- Q2.** The cost function typically minimized in linear regression is:
(a) Cross-entropy (b) Hinge loss (c) Mean Squared Error (d) Gini index
- Q3.** Logistic regression outputs values in the range:
(a) $(-\infty, \infty)$ (b) $[0, 1]$ (c) $[-1, 1]$ (d) $\{0, 1\}$
- Q4.** A correlation coefficient of -0.95 indicates:
(a) Weak positive (b) No relationship (c) Strong negative (d) Perfect positive
- Q5.** In a confusion matrix, a False Positive (Type I error) is:
(a) Correctly predicted positive (b) Predicted positive, actually negative (c) Predicted negative, actually positive (d) Correctly predicted negative
- Q6.** Stochastic Gradient Descent updates model weights using:
(a) The entire dataset (b) A single sample (or small batch) per step (c) Only the last epoch (d) No data
- Q7.** Which metric is most reliable for an imbalanced classification dataset?
(a) Accuracy (b) F1-Score (c) Sum of weights (d) Learning rate
- Q8.** Stratified sampling is used to:
(a) Pick every k-th record (b) Preserve class proportions in the sample (c) Remove outliers (d) Increase the learning rate
- Q9.** Overfitting is best described as a model that:
(a) Performs poorly on both train and test (b) Performs well on train but poorly on test (c) Ignores the training data (d) Has high bias
- Q10.** The learning rate in gradient descent controls:
(a) The number of features (b) The step size of each weight update (c) The dataset size (d) The number of classes

Section B — Short Answer Questions

(Answer any six · 6 × 3 = 18 Marks)

Q11. Define machine learning. Differentiate between supervised and unsupervised learning with one example each. [3]

Q12. What is the difference between correlation and causation? Give an example. [3]

Q13. Explain the terms Bias and Variance. What is the bias–variance trade-off? [3]

Q14. State the difference between Batch Gradient Descent and Stochastic Gradient Descent (SGD). [3]

Q15. Define Precision, Recall, and F1-Score using the confusion matrix. [3]

Q16. What is the sigmoid function? Write its equation and state why it is used in logistic regression. [3]

Q17. Distinguish between random sampling and stratified sampling. [3]

Q18. What is feature scaling? Name two common techniques. [3]

Section C — Long Answer Questions

(Answer any four · 4 × 8 = 32 Marks)

Q19. Explain Simple Linear Regression. Derive the cost function (MSE) and describe how the parameters (slope and intercept) are estimated. Illustrate with the line equation $y = mx + c$. [8]

Q20. Describe Logistic Regression for binary classification. Explain the hypothesis function, the decision boundary, and the log-loss (cross-entropy) cost function. [8]

Q21. Explain the Gradient Descent optimization algorithm. Contrast Batch, Stochastic, and Mini-batch Gradient Descent, listing one advantage and one disadvantage of each. Discuss the role of the learning rate. [8]

Q22. What is a confusion matrix? For a binary classifier, define TP, TN, FP, and FN, then derive Accuracy, Precision, Recall, Specificity, and F1-Score. Explain why accuracy alone can be misleading on imbalanced data. [8]

Q23. Discuss common sampling techniques in machine learning — simple random, stratified, systematic, and cluster sampling. State when each is appropriate. [8]

Q24. Explain correlation analysis. Define the Pearson correlation coefficient, its range and interpretation, and describe how a correlation matrix / heatmap helps in feature selection. [8]

Section D — Numerical / Practical Problems

(Answer any two · 2 × 5 = 10 Marks)

Q25. A binary classifier produces the following results on 100 samples: TP = 40, TN = 30, FP = 20, FN = 10. Compute Accuracy, Precision, Recall, and F1-Score. [5]

Q26. For the dataset $X = [1, 2, 3, 4, 5]$ and $Y = [2, 4, 5, 4, 5]$, compute the Pearson correlation coefficient between X and Y (show intermediate sums). [5]

Q27. Given a single training example $x = 2$, $y = 3$ with current parameters $w = 0.5$, $b = 0$ and learning rate $\alpha = 0.1$, perform ONE step of SGD for linear regression (prediction $\hat{y} = wx + b$, squared-error loss) and report the updated w and b . [5]